

ML Model Selection — Step-by-Step Decision Guide

Goal: Dataset dekhkar systematically decide karna ki kaunse ML models try karne chahiye. Ye rigid rule nahi hai; final choice validation/CV performance + problem requirements par depend karti hai.

STEP 1 — Target identify karo

Continuous numeric target → Regression
Examples: price, salary, temperature.

Category/class target → Classification
Examples: spam/not spam, pass/fail, disease/no disease.

STEP 2 — Data size check karo

Situation	Good first candidates / notes
Small dataset	Linear/Logistic Regression, SVM, KNN, Decision Tree; CV zaroor use karo.
Medium tabular	Random Forest, Gradient Boosting, XGBoost + simple baseline.
Large tabular	Logistic/Linear baseline; tree boosting (especially XGBoost) often strong; compute cost check karo.
Very high-dimensional / sparse text	Naive Bayes or Logistic Regression/SVM are strong candidates.

STEP 3 — Relationship / boundary dekho

Approximately linear relationship → Linear Regression / Logistic Regression.

Clearly nonlinear / interactions → Decision Tree, Random Forest, Gradient Boosting, XGBoost; classification me SVM (kernel) bhi candidate.

Nearby points ka behavior important → KNN candidate; scaling important.

STEP 4 — Feature type aur preprocessing

Data condition	What to consider
Numerical features, different scales	KNN and SVM ke liye StandardScaler/appropriate scaling important. Logistic/Linear bhi scaling se benefit kar
Categorical features	One-hot/appropriate encoding; tree boosting ke implementation ke according handling.
Text / word counts / TF-IDF	Naive Bayes, Logistic Regression, Linear SVM strong starting candidates.
Missing values	Impute/handle first; XGBoost missing values handle kar sakta hai, but preprocessing still dataset-dependent.
Outliers	Linear/KNN/SVM sensitive ho sakte hain; tree-based models generally more robust.

STEP 5 — Explainability / requirement check

Need	Model candidates
Very interpretable coefficients	Linear Regression / Logistic Regression
Readable if-then rules	Decision Tree
Strong tabular performance	Random Forest / Gradient Boosting / XGBoost
Text classification baseline	Naive Bayes / Logistic Regression / Linear SVM
Small/medium data + margin-based separation	SVM
Local similarity matters	KNN

STEP 6 — Build a baseline first

Regression: start with Linear Regression (when appropriate).

Classification: start with Logistic Regression (when appropriate).

Baseline ka purpose: complex model se pehle ek reference performance milna.

STEP 7 — Candidate models choose karo

Regression: Linear Regression → Decision Tree Regressor → Random Forest Regressor → Gradient Boosting/XGBoost.

Classification: Logistic Regression → KNN/SVM/Decision Tree → Random Forest → Gradient Boosting/XGBoost.

Text/sparse data me Naive Bayes/Logistic/SVM ko priority do.

STEP 8 — Validation + tuning

Train/test split → preprocessing pipeline → Cross-Validation → GridSearchCV/RandomizedSearchCV → compare CV metrics → final test evaluation.

Important: Test set ko tuning ke liye baar-baar use mat karo. Final test evaluation end me karo.

STEP 9 — Metric problem ke according choose karo

Problem	Metrics
Regression	MAE, MSE, RMSE, R^2
Balanced classification	Accuracy + Precision/Recall/F1 as relevant
Imbalanced classification	Precision, Recall, F1, PR-AUC/ROC-AUC as appropriate; accuracy alone avoid when misleading.
False negative costly	Recall ko priority do.
False positive costly	Precision ko priority do.

Quick Decision Tree

1. Target continuous? → Regression. Else → Classification.
2. Need simple/interpretable model? → Linear/Logistic or Decision Tree.
3. Text/sparse features? → Naive Bayes / Logistic / Linear SVM.
4. Small/medium scaled dataset + strong boundary? → SVM/KNN candidates.
5. Nonlinear tabular data? → Random Forest / Gradient Boosting / XGBoost.
6. Need strong tabular performance? → include XGBoost/boosting candidates.
7. Compare using Cross-Validation and the right metric; tune only after selecting sensible candidates.

Golden Rule

Algorithm selection is not: “one dataset → one fixed model.”

It is: Problem → Data → Preprocessing → Baseline → Candidate models → Cross-validation → Tuning → Metric-based comparison → Final test.